

MEASURING DAYS

SOLAR TIME

Solar time is the way of gauging time from the Sun's apparent motion across the sky, as measured by a sundial. One solar day is subdivided into 24 hours.

Every day, Earth rotates once, and most locations on its surface pass from sunlight to shadow and back, producing the day–night cycle. However, there are two possible definitions for what constitutes a day, and only one of these, the solar day, lasts for exactly 24 hours. A solar day is defined by the apparent movement of the Sun across the sky produced by Earth's rotation. It is the length of time the Sun takes to return to its highest point in the sky from the same point the previous day. The other type of day, the sidereal day, is defined by Earth's rotation relative to the stars. It is the length of time a star takes to return to its highest point in the sky on successive days. A sidereal day is 4 minutes shorter than a solar day.



APRIL 1, 8:00 PM.



APRIL 8, 8:00 PM.



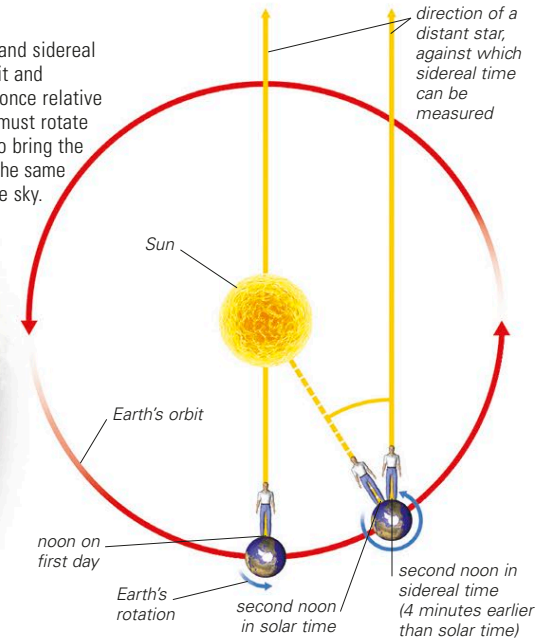
APRIL 15, 8:00 PM.

SIDEREAL TIME

The distinctive constellation Orion (see pp.390–391), here pictured as if from 50°N, appears lower in the sky at the same solar time each day, as the daily 4-minute difference between solar and sidereal time mounts up.

SOLAR AND SIDEREAL DAY

The disparity between solar and sidereal days results from Earth's orbit and rotation. After rotating once relative to the stars, Earth must rotate a little farther to bring the Sun back to the same point in the sky.



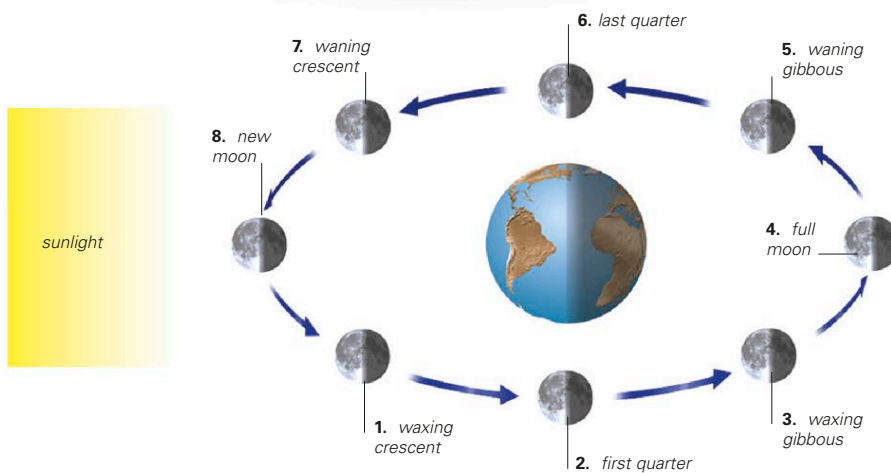
MEASURING MONTHS

The concept of a month is based on the Moon's orbit around Earth. During each of the Moon's orbits, the angle between Earth, the Moon, and the Sun continuously changes, giving rise to the Moon's phases. The phases cycle through new moon (when the Moon is between Earth and the Sun), crescent, quarter, and gibbous, to full moon (when the Earth lies between the Moon and the Sun). A complete cycle of the Moon's phases takes 29.5 solar days and defines a lunar month. However, Earth's progress around the Sun complicates the expression of a month, just as it confuses the measurement of a day. The Moon in fact takes only 27.3 days to orbit Earth with reference to the background stars. Astronomers call this period a sidereal month. The disparity results because Earth's progress around the Sun

alters the angles between the Earth, Sun, and Moon. After one full orbit of Earth (a sidereal month), the Moon must orbit a bit farther to return to its original alignment with Earth and the Sun (a lunar month).

CHANGING ANGLES

During each lunar orbit, the angle between Earth, the Moon, and the Sun changes. The part of the Moon's sunlit face seen by an observer on Earth changes in a cyclical fashion.



1. WAXING CRESCENT 2. FIRST QUARTER 3. WAXING GIBBOUS 4. FULL MOON 5. WANING GIBBOUS 6. LAST QUARTER 7. WANING CRESCENT 8. NEW MOON